

```

    } catch (GatewayException ex) {
    } catch (SMSLibException ex) {
    } catch (IOException ex) {
    } catch (InterruptedException ex) {
    }
}

```

Sample code to receive an SMS is:

```

SerialModemGateway gateway = new SerialModemGateway("Huawei",
"COM8", 115200, "", "");
gateway.setProtocol(Protocols.PDU);
gateway.setInbound(true);
gateway.setOutbound(true);
Service.getInstance().setInboundMessageNotification
(inboundNotification);
Service.getInstance().addGateway(gateway);
Service.getInstance().startService();
msgList = new ArrayList<InboundMessage>();
Service.getInstance().readMessages(msgList,
MessageClasses.ALL);

for (InboundMessage msg : msgList)
    System.out.println(msg);

System.out.println("Waiting for SMS. Press any key to stop.");
System.in.read();

/*Class to display the notification and text of incoming
message*/

public class InboundNotification implements
IInboundMessageNotification
{
    public void process(AGateway gateway, MessageTypes msgType,
InboundMessage msg)
    {
        if (msgType == MessageTypes.
INBOUND)
        {
            System.out.println("Incoming message from: " + gateway.
getGatewayId());
            System.out.println("Sender: "+ msg.
getOriginator());
        }

        System.out.println(msg);
    }
}

```

The entire code for the above mentioned snippets is available at <http://goo.gl/uQDsCO> as a NetBeans project.

The DynamicReports API

I came across the DynamicReports API when I was

developing my first business application. I was stuck for a while on the reporting module, thinking that I'd need to do it from scratch. That wasn't a motivating thought at all, so I searched the Internet for an alternative. I came across several proprietary reporting APIs before I stumbled upon DynamicReports, and since then I haven't looked anywhere else to create reports.

The DynamicReports API, developed by Ricardo Mariaca, is based on Jasper Reports. It reduces the effort required to create reports programmatically by abstracting much of the work required when using the Jasper Reports API. It comes pre-built with several designs and templates, thus eliminating the need for a visual designer. It also allows you to include various charts (bar charts, pie charts, metre charts, etc) in reports. It also allows you to export the reports in various formats (.pdf, .xls, .doc, etc). Thus, apart from some basic formatting and source data, it takes care of pretty much everything. You can visit <http://www.dynamicreports.org> to read more about it. It is well documented and the community provides good support.

Download and implementation

You can download the API from <http://goo.gl/9xwY34>. At the time of writing, the version is 3.2.1. Once the zip file is downloaded, extract it and copy the contents of the *dist* folder to your project. Next, import the JAR files in your NetBeans project using the steps mentioned previously.

DynamicReports hosts a variety of examples at <http://www.dynamicreports.org/examples> to get you started. In case you need help, you can visit the support forum at <http://www.dynamicreports.org/forum/> and post your queries there or contact me at the email ID given at the end of the article.

The above mentioned APIs can be used in a variety of applications, either alone or in combination. For example, almost any application that requires users to input some value can benefit from the SimpleValidation API. DynamicReports can enhance any report-intensive business application (MIS, ERP, school results management, etc). SMSLib is very convenient for applications that use SMS as a means to accept user input (like accepting orders via SMS), and applications designed for machine-to-machine communication (for example, applications to monitor radio taxis, meter reading applications, etc).

Open source APIs and their developers are a boon to the developer community. They not only help in reducing development time and efforts but also bring down the overall cost of the project. 

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